



Faculty of Science and Technology

Programme Structure & Syllabus

For

First Year

Bachelor of Science (Honors / Research)

Computer Science

(BSC CS)

P-2024

With Effect from Academic Year 2024– 2025

	Authority	Date
Proposed by	Board of Studies in Computer Science	
Approved by	Academic Council, Vishwakarma University, Pune	

Issued by

Chairman – BoS

Dean of Faculty

Director, IQAC

Title: Programme Structure and Syllabus

Form No: IQAC-101

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the University	
Emerge as a Premier University Recognized Internationally for Excellence in Education, Research and Innovation	
Mission of the University	
VU-M1	To impart contemporary transformative education through research and innovation
VU-M2	To develop competent leaders-professionals for life and livelihood
VU-M3	To co-create human and socio-economic capital par excellence
VU-M4	To inculcate life skills and holistic culture appreciating morals and ethics
Values of the University	
Excellence	Transparency
Innovation	Sustainability
Diversity	Responsibility
Adaptability	Compassion

Vision of the Department of B.Sc. (Computer Science)	
Emerge as a renowned research and innovation-oriented department in the field of Information Technology.	
Mission of the Department of B.Sc. (Computer Science)	
M1	To impart industry centric education through research and innovation
M2	To develop skilled professionals to meet global socio-economic challenges
M3	To introduce best practices in software development adhering to moral and ethics.
M4	To foster collaborative approach and leadership skills.
Values of the Department of B.Sc. (Computer Science)	
Excellence	Transparency
Innovation	Sustainability

Diversity	Responsibility
Adaptability	Compassion

Programme Educational Objectives (PEOs)

PEO No.	Statement
PEO1	Graduates will have a thorough foundational knowledge of the principles and practices of Computer Science, and be proficient in applying them.
PEO2	To be able to adapt new tools/techniques and effectively contribute as an individual and lead the software development team to provide economically and technically viable solutions to a problem.
PEO3	Commit to ethical and moral practices for providing solutions to societal problems through life-long learning and research.

Mapping of Mission Statement of Department to PEOs

Mission Statement	PEO1	PEO2	PEO3
M1	3	2	3
M2	1	3	2
M3	1	2	3
M4	1	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Outcomes (POs)

PO No.	Statement
PO1	Possess profound knowledge of theory and principles of basic mathematics, statistical techniques and algorithms in the field of computer science.
PO2	Ability to think critically and select economically and technically viable solutions to a given problem.
PO3	Ability to design, develop and deploy solutions for identified problems from diverse domains.
PO4	Provide proof of concept and present novel solutions to challenges encountered by industry.

PO5	Adopt new programming languages, tools , and processes followed in industry
PO6	To work independently or as an effective team member of a software development project.
PO7	Effectively communicate with diverse clients and stakeholders for software project management.
PO8	Interpersonal skills to interact with people from diverse cultural backgrounds
PO9	Adhere to the professional, ethical and legal practices followed in the IT industry.
PO10	Possess software project management skills like: cost estimation, effort estimation, scheduling, and team building.

Mapping of PEOs to POs

PEO Number	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
PEO1	3	2	3	2	1	-	-	-	-	-
PEO2	1	3	-	2	3	2	2	1	1	2
PEO3	-	-	2	-	-	1	1	2	3	1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Programme Specific Outcomes (PSOs)

PSO No.	Statement
PSO1	Choose appropriate data structure, algorithm, and software development paradigm to address real life problems.
PSO2	Apply modern tools and techniques used in Data Science and Game Development.
PSO3	Provide the solutions to computing challenges from multidisciplinary fields.
PSO4	Possess a research and development attitude focusing on knowledge creation and innovation.

Mapping of PEOs to PSOs

PEO Number	PSO1	PSO2	PSO3
PEO1	3	1	2
PEO2	2	3	2
PEO3	1	2	3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial

Program Structure

Semester –I									
Cours e Type	Course Code	Course Name	Teaching Scheme (Hours/Week)			Credit	Examination Scheme and Marks		
			L	T	P	C	CIE	ESE	Total
DSC	BSCCS24101	Problem Solving and Programming	3	0	0	3	60	40	100
DSC	BSCCS24102	Problem Solving and Programming Lab	0	0	2	1	15	10	25
CC	BSCCS24103	Capstone Project-I	0	0	4	2	50	0	50
MC	MT24CS01A	VU Level - Principles of Computers and Information Technology	2	1	0	3	60	40	100
OE	OE24CS01A	VU Level - Computer Network	2	1	0	3	60	40	100
SEC	SE24CS01A	VU Level- Basic Computing	1	1	0	2	50	0	50
AEC	AE24CS01A	VU Level- English	2	0	4	4	60	40	100
VEC	VE24CS01A	VU Level - Digital Technology Solutions	1	1	0	2	50	0	50
TOTAL			11	4	10	20	405	170	575
Instructions, if any: 1 Theory or 1 Tutorial Hour = 1 Credit, 2 Practical hours = 1 Credit									

Name of Head of Department : Dr. Rajkumar Jagdale

Name of Dean: Dr. Nitin Satpute

Sign:

Sign:

Date:

Date:

BSCCS24101:: Problem Solving and Programming

Course Type	Department Specific Core Course	Semester	I
-------------	---------------------------------	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	03 Hr./Week	Lecture	3	CIE Marks	60
Tutorial	00 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	02 Hr./Week	Practical/Studio	1	Total Marks	100
Total	05 Hr./Week	Total	4		

Course Description

The objective of this course is to introduce various concepts of programming to the students using the C programming language.

Course Outcomes

CO No.	Statement
1	Summarize the concepts of programming before actually starting to write new programs
2	Comprehend what happens in the background when the programs are executed
3	Develop logic for Problem Solving.
4	Develop basic constructs of programming such as data, operations, conditions, loops, functions etc.
5	Apply the problem-solving skills using syntactically simple language i. e. C programming language

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	1	-	-	-	-	-	-	-	-	-	1	-	-	1	1, 2,

CO2	1	-	-	-	-	-	-	-	-	-	1	-	-	1	1, 4
CO3	-	-	3	-	-	-	-	-	-	-	-	-	3	1	1, 3, 5
CO4	-	-		-	-	-	-	2	-	-	-	-		1	1, 3, 5
CO5	-	-	3	-	-	-	2	-	-	-	-	-	3	1	1, 3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No – I	Title of Unit - Introduction and C Fundamentals	Hours – 09	CO	BTL
Basics of Computer Languages: Machine language, Assembly Language, High level languages, Compiler, Interpreter, Linker. Problem Statement, Algorithm, Flowchart, History of C, Character set of C, Identifiers and Keywords, Variable and Constants, Defining constants using const and #define, Data Types, Type modifiers, Operator and its types, Operator precedence, Expressions, Structure of a „C“ Program, Preprocessor directives (# include and #define) , Escape sequence characters, Input and Output Functions: printf() and scanf(), getchar(), putchar(), gets(), puts ()			1	2
Unit No – II	Title of Unit – Control Statements	Hours – 09	2	1
Branching – Simple if, if ... else, nested if...else, else-if ladder, switch-case, Looping – while, do...while, for loop, Nested Loops, Loop interruption statements: break, continue, goto statement.				
Unit No – III	Title of Unit – Functions	Hours – 09	3	4
Introduction, Types of functions, Library and user defined function, Function declaration/prototype, function definition, Arguments: Formal and Actual, Call by value, Recursion, Command line arguments. Storage classes: automatic (internal/local), global (external), static and register.				
Unit No – IV	Title of Unit – Arrays	Hours – 09	4	3
Declaration, Memory representation, One and two dimensional array, Initialization, Character Array, Strings. Passing arrays to function, String handling functions.				
Unit No – V	Title of Unit – Pointers & Structures	Hours – 09	5	5
Pointer: Declaration and Initialization, address of (&) and indirection (*) operator, Pointer and arrays, Pointer arithmetic, Pointer to pointer, Passing pointers to function. Call by reference. Structures: Creating structures, Defining Structure variable, Accessing members of structures (dot operator), Initialization of structures, Array of structures.				
Practical				
Group A				
1. Structure of a C Program				

2. Write programs in C which implements basic C operators 3. Write programs in C which implements concept of conditional construct (if-else) 4. Write programs in C which implements concept of conditional construct (switch-case) 5. Write programs in C which implements concept of for loop 6. Write programs in C which implements concept of nested for loop 7. Write programs in C which implements concept of while/do...while loop 8. Write programs in C which implements concept of pointer Group B 1. Write a program in C which implements concept of array 2. Write a program in C which implements concept of structure 3. Write a program in C which implements the concept of function. 4. Write a program in C to carry out following operations on strings using library functions a. To concatenate a string S2 to string S1. b. To find the length of a given string c. To compare two strings S1 and S2. d. To copy a string S2 to another string S1.		
---	--	--

Textbooks

1	Kernighan, Ritchie, "The C Programming Language", Prentice Hall of India
2	Behrouz A. Forouzan, Richard F. Gilberg, "Computer Science: A Structured Programming Approach Using C", Cengage Learning © 2007
3	Yashavant Kanetkar, "Let Us C", BPB Publications

Reference Books / Journal Articles / Weblink

1	Pradeep K. Sinha and Priti Sinha, "Computer Fundamentals", BPB Publications
2	"Programming approach using C", Indian Edition, Thomson, 3rd edition
3	Carlo Ghezzi, Mehdi Jazayeri, "Programming Language Concepts", John Wiley and Sons
4	E. Balagurusamy, "Programming in ANSIC C", Tata McGraw Hill, 2002
5	Richard Petersen, "The Complete Reference Linux", McGraw-Hill Publications

MT24CS01A:: Principles of Computers and Information Technology

Course Type	Minor Course	Semester	I
-------------	--------------	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	02 Hr./Week	Lecture	2	CIE Marks	60
Tutorial	01 Hr./Week	Tutorial	1	ESE Marks	40
Practical/Studio	00 Hr./Week	Practical/Studio	0	Total Marks	100
Total	03 Hr./Week	Total	3		

Course Description

The objective of this course is to introduce basic concepts and principles of Information Technology with illustrations of basic concepts of information technology, Computer basics, Internet concepts and computer networking.

Course Outcomes

CO No.	Statement
1	Understand the basic concepts of information technology
2	Understand the basics of computers.
3	Learn types of computer networking
4	Learn internet concepts and communication modes
5	Understand internet addressing

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	1	2	2						3		1	1	1, 2,
CO2		1	1	2	2						3		1	1	1, 4

CO3		1	1	2	2					3		1	1	1, 3, 5
CO4		1	1	2	2					3		1	1	1, 3, 5
CO5		1	1	2	2				2	3		1	1	1, 3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No: 1	Title of Unit: Introduction to Information Technology	09 Hours	CO	BTL
Information Technology: Introduction – Information systems – Definition of computer and system – Software and Data - IT in business and Industry – IT in the Home and at Play – It in Education and Training – IT in Entertainment and the Arts – IT in Science, Engineering, and Mathematics – Global Positioning System(GPS).			1	1, 2
Unit No: 2	Title of Unit: Introduction to Computer	09 Hours	2	2, 4
Introduction to Computers, History of computers, Types of computers, Characteristics of computers, Basic Anatomy of a computer, Applications of computer – Memory – Memory types. Software- Kinds of Software - The five types of Applications software - Word processing – Spreadsheets - Database software, Presentation graphics software - Communications software- System Software – Operating system - functions.				
Unit No: 3	Title of Unit: Computer Networks	09 Hours	3	3, 4
Computer Networks: Introduction – Definition Computer Networks - Types of Networks – Local Area Network – Metropolitan Area Network - Wide Area Network – Personal Area Network - internet – Intranet – firewalls - Network Topology – Bus – Ring – Hybrid – Star.				
Unit No: 4	Title of Unit: Internet Concepts	09 Hours	4	2, 3
Basic Internet Concepts: – Analog and Digital Signals - modems and communication Software, ISDN lines, and Cable Modems - Definition of Internet - The World Wide Web - Connecting to the Internet – Browsing the web – Web browser – Uniform Resource Locator (URL) – E-mail communication.				
Unit No: 5	Title of Unit: Internet Addressing	09 Hours	5	1, 2,3
Internet address - Domain Name System (DNS) – Locating information on the net – Internet Search Engines – Chatting and conferencing on the Internet Online Chatting –Messaging – Usenet Newsgroup.				

Textbooks

1	Introduction To Information Technology, V. Rajaraman, PHI
2	Data Communications & Networking -Behrouz Ferouzan (III Edition), McGraw-Hill publications, 4 th edition, 2014

Reference Books / Journal Articles / Weblink

1	https://www.techtarget.com/searchdatacenter/definition/IT
2	javatpoint.com/types-of-computer-network

OE24CS01A :: Computer Networks

Course Type	Open Elective	Semester	I
-------------	---------------	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hrs./Week	Lecture	2	CIE Marks	60
Tutorial	1 Hrs./Week	Tutorial	1	ESE Marks	40
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	100
Total	3 Hrs./Week	Total	3		

Course Objectives :

1. Understand the basic concepts of computer networks.
2. Types of networks, devices, transmission Basics & Networking Connectivity Devices
3. Understand OSI model, TCP IP Model
4. Understand protocols and concept of Network Security
5. Understand basics of cloud computing

Course Outcomes

CO No.	At the end of this course students will be able to
1	Understand the basic functions in computer networks.
2	Understand the basics of protocols, devices, transmission medias
3	Understand OSI model, TCP IP Model
4	Understand protocols and concept of Network Security
5	Understand basics of cloud computing

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	2								1				1
CO2		2	1	1	1									1	4
CO3		1	1	1	1										2
CO4		1	1	1								1			2
CO5	1	1	1	1									2		1

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No: 1	Title of Unit: Basics of Computer Networks	12 hrs	CO	BTL
Definition of Computer Network, Basic Terminologies of Computer Networks, Types of Networks – Local Area Network – Metropolitan Area Network - Wide Area Network – Personal Area Network, Topology & types of topology, Communication Types, Modes of Communication, Server Based LANs & Peer-to-Peer LANs, standards			1,2	2
Unit No: 2	Title of Unit : Transmission Basics & Networking Connectivity Devices	10 hrs	2	
Guided Media(Wired), Propagation Methods, Unguided Media(Wireless), Connectivity Devices, internet – Intranet – extranet ,firewalls, Network connectivity Devices				2
Unit No: 3	Title of Unit: Network Models & Addressing	10 hrs		
ISO-OSI Reference Model, Internet Model (TCP/IP), Comparison of ISO-OSI & TCP/IP Model, Addressing, IP Addressing, circuit switching, packet switching			3	3
Unit No: 4	Title of Unit : Protocols and Network Security	5 hrs		
What is protocols, study of various protocols at each layer , Introduction to the Concepts of Security, The Need for Security, concepts of Cryptographic, Encryption and Decryption			4	2
Unit No: 5	Title of Unit : Basics of cloud computing	8 hrs		
The Vision of Cloud Computing, Defining a Cloud, Applications and Benefits of cloud computing, Storage as a service, Database as a service, Process as a service, Information as a service			5	3

Textbooks

1	Data Communications & Networking -Behrouz Ferouzan (III Edition), McGraw-Hill publications, 4 th edition, 2014
2	Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, S.Thamarai Selvi from TMH 2013.
3	Kahate, A. (2003). Cryptography and Network Security. McGraw-Hill Education

Reference Books / Journal Articles / Weblink

Reference Books	
1	Computer Networks by Andrew Tanenbaum, Pearson Education, 4thEdition, PHI publications.
2	Networking All In One Dummies Wiley Publication, 5thEdition, 2012
3	Stallings, W. (2016). Cryptography and Network Security: Principles and Practice (7th ed.). Pearson.

SE24CS01A:: Basic Computing

Course Type	Skill Enhancement Courses	Semester	I
-------------	---------------------------	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hrs./Week	Lecture	1	CIE Marks	50
Tutorial	1 Hrs./Week	Tutorial	1	ESE Marks	-
Practical/Studio	0 Hrs./Week	Practical/Studio	0	Total Marks	50
Total	3 Hrs./Week	Total	2		

Course Objectives

The aim of the course is to generate the manpower with adequate theory and practical knowledge of Basic Computer, operating systems, Word Processing and Word Processors, Spreadsheets and presentations

Course Outcomes

CO No.	At the end of this course students will be able to
1	Understand Computer working and operating system
2	Use Word Processors for documentation
3	Use Spreadsheets to analyze data using formulas and charts
4	Create slides for presentations

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BTL
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1		1	1		1						1		1	1	1,2
CO2		1	1		2						1		1	1	2,3
CO3		1	1		2						1		1	1	2,3
CO4		1	1		2						1		1	1	3,4

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL:Bloom's Taxonomy Level

Course Content

Unit No.: 1	Title of Unit : Introduction to Computer	3 Hours	CO	BTL
Introduction to Computer, Hardware, Software, Input Output Devices, Memory			1	2
Unit No.: 2	Title of Unit: Operating System	3 Hours		
Operating system definition, Windows operating systems and its features, Introduction to icons and features, start menu , control panel , organize and manage files and folders			2	2
Unit No.: 3	Title of Unit : Word Processing and Word Processors	8 Hours		
Introduction to word processing and word processors, Examples of word processors: MS Word, Google docs, Features of MS Word, Create, Open, Edit, Save, Close, Print document , Format Text, Paragraphs, and Section, Create Tables and Lists, Insert and Format Graphic Element			2,3	3
Unit No.: 4	Title of Unit : Spreadsheet	10 Hours		
Introduction to Spreadsheet, Examples of Spreadsheet MS Excel, Google sheet, Features of Excel, Create, Open, Edit, Save, Close, Print spreadsheet, Formatting spreadsheet, Creating Formulas, Copying and pasting formulas, Selecting, hiding and unhiding,, inserting and deleting rows and columns, Adjusting column width and row height, working with charts, pivot table			4	2
Unit No.: 5	Title of Unit : Presentation	6 Hours		
Introduction to Presentation, Examples of presentation MS PowerPoint, Google slides, Create, Open, Edit, Save, Close, Print presentation, Applying themes to slides, working with objects, Slide transitions, Animation			5	3

Textbooks

1	Introduction To Information Technology, V. Rajaraman, PHI
2	Microsoft Word, Excel, and PowerPoint: Just for Beginners, Dorothy House

Reference Books / Journal Articles / Weblink

Reference Books	
1	Data Communications & Networking -Behrouz Ferouzan (III Edition), McGraw-Hill publications, 4th edition, 2014
2	https://www.udemy.com/course/learn-microsoft-excel-powerpoint-and-word/
3	www.javatpoint.com

VE24CS01A:: Digital Technology Solutions

Course Type	Value Education Courses	Semester	I
-------------	-------------------------	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	1 Hr./Week	Lecture	1	CIE Marks	50
Tutorial	1 Hr./Week	Tutorial	1	ESE Marks	0
Practical/Studio	0 Hr./Week	Practical/Studio	0	Total Marks	50
Total	2 Hr./Week	Total	2		

Course Description

This course covers digital technology solutions for business covering use of Tableau. Tableau Creator functionality required for new Tableau users. It starts with simple visualizations and moves to an in-depth look at the different chart and graph functions, calculations, mapping and other functionality. The lessons are organized into units that provide conceptual information, how-to steps, and tutorials. The course provides the opportunity to visually analyze data, build Interactive dashboards and share them with others. Students will create and design both static and dynamic tables, data visualizations, dashboards, and stories while incorporating visual design best practices to better communicate insights to the intended audience, such as business stakeholders.

Course Outcomes

CO No.	Statement
1	To understand how Digital Technology solutions help business growth.
2	To understand the value, structure and architecture of Tableau Software as it applies to data visualization in the industry of business and data analytics.
3	To analyze and build interactive tables by connecting, preparing, and customizing data in Tableau.
4	To create and evaluate data visualizations, dashboards, and Tableau Stories, to communicate analytic insights to the intended audience, such as business stakeholders.
5	Apply Tableau performance optimization to improve the speed of working with large data.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs				BT L
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PSO 1	PSO 2	PSO 3	PSO 4	
CO1	2				2							1			1
CO2		3									1				2
CO3			3								2		1		4
CO4							3	1			1		2		5
CO5													3		3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No.: 1	Title of Unit: Digital technology solutions	Hours: 4	CO	BTL
Definition Digital technology solutions, Need of Digital technology solutions in businesses, Impact areas in business and Digital technology solutions, Importance of Tableau in business. Tableau Use cases			1, 2	1, 2
Unit No.: 2	Title of Unit: Introduction to Data Visualization and The Power of Tableau and Architecture of Tableau	Hours: 10	1, 2	1, 2
What is data visualization?, Comparison and benefits against reading raw numbers Installing Tableau, Tableau interface Architecture of Tableau, Interface of Tableau (Layout, Toolbars, Data Pane, Analytics Pane, etc.), Tableau field types, Saving and publishing a data source, Live vs extract connection, Various file types, The ways to share and export the work done in Tableau				
Unit No.: 3	Title of Unit: Charts and Graphs	Hours: 4	3	1, 3
Basic Charts – Histograms, Box plot, Pie, Bar, Line, Bubble, Scatter, Tree, Maps, Text table. Dual axes graphs – Waterfall charts, Market basket analysis (MBA)				
Unit No.: 4	Title of Unit: Working with Metadata and Data Blending	Hours: 6	4, 5	4, 5
Connect to Excel sheet to import data, Use metadata and extracts, Manage NULL values, Clean up data before using, Perform the join techniques, Execute data blending from multiple sources				
Unit No.: 5	Title of Unit: Hands on Tableau	Hours: 6	4, 5	4, 5
Play with Tableau desktop, Learn about the interface, Share and export existing works Plot a histogram, tree map, heat map, funnel chart, and more using the given dataset. Perform market basket analysis (MBA) on the same dataset				

Textbooks

1	Tableau 10 for Beginners: Step by Step Guide to Developing Visualizations in Tableau 10, Chandraish Sinha, Createspace Independent Pub, 2017
2	Practical Tableau: 100 Tips, Tutorials, and Strategies from a Tableau Zen Master, Ryan Sleeper, O'Reilly Media; 1st edition, 2018

Reference Books / Journal Articles / Weblink

1	Visual Analytics with Tableau, Alexander Loth, Nate Vogel, Sophie Sparkes, Wiley; 1st edition (May 7, 2019)
2	Tableau Your Data!: Fast and Easy Visual Analysis with Tableau Software, Daniel G. Murray, Wiley; 2nd edition (January 26, 2016)
3	Learning Tableau 2020: Create effective data visualizations, build interactive visual analytics, and transform your organization, Joshua N. Milligan, Packt Publishing; 4th ed. edition (August 31, 2020)

AE24CS01A:: English

Course Type	AEC	Semester	I
-------------	-----	----------	---

Teaching Scheme		Credits		Examination Scheme	
Lecture	2 Hr./Week	Lecture	2	CIE Marks	60
Tutorial	0 Hr./Week	Tutorial	0	ESE Marks	40
Practical/Studio	4 Hr./Week	Practical/Studio	2	Total Marks	100
Total	6 Hr./Week	Total	4		

Course Description

This rigorous course targets to imbibe basic and advanced English Language skills among undergraduate students. Students will get a robust understanding of various essential concepts in English Language Usage from a practical point of view. This course will help to bring crucial professional development in the behaviour of the students, which will enable them to cope with social, formal and organizational situations.

Course Outcomes

CO No.	Statement
1	Students can contextualize knowledge of the English language for practical purposes.
2	The norms of language will be acquired naturally by the students and infer language structure and its usage.
3	The essential language etiquettes such as body language, tone, and group behaviour will be imbibed among the students.
4	Students will be able to exhibit strong presentation abilities.
5	Students decode the mechanism of language and use it for successful accomplishments in competitive examinations and employability-related speaking and writing tasks.

Mapping of COs to POs and PSOs

CO Number	POs										PSOs			PSO 4	BTL
	PO 1	PO 2	PO 3	PO 4	PO5	PO 6	PO 7	PO 8	PO 9	PO 10	PS O1	PS O2	PSO 3		
CO1							2	2						2	1
CO2	2		2								1		2	2	2
CO3							3	3						2	3
CO4						1	1				1	2			1
CO5		2		2					2	2	2	2			2,3

Affinity Level: 1- Slight, 2- Moderate, 3-Substantial, BTL: Bloom's Taxonomy Level

Course Content

Unit No.: 1	Language and Usage	Hours 15	CO	BTL
1. Language and Society 2. Communication process, channel and models 3. Types/ flow of communication 4. Barriers to communication and strategies to overcome.: Case studies			1,2	1,3
Unit No.: 2	Listening & Reading Skills	Hours 10		
1. Distinction between listening and hearing 2. Types of Listening 3. Barriers to Listening 4. Roles and responsibilities of the listener toward the speaker 5. Cultivating empathetic listening 6. IELTS Listening practices 7. International Phonetic Alphabets (IPA) (Phonetic Chart) 8. Pitch variation 9. AmE and BrE (Words, spellings, pronunciations and phrasal verbs) 10. Types of reading/ Skimming, Scanning and intensive				
Unit No.: 3	Vocabulary Building	Hours 10		
1. Synonyms, antonyms, homonyms, homophones, etc. 2. Semantics, syntax, deep structure, surface structure, etc. 3. Collocations 4. Idioms and Phrases 5. Word formation (affixation, conversion, clipping, blending, compounding, etc) 6. Power words			3,4	3,5,6
Unit No.: 4	Speaking Skills	Hours 10		
1. Speaking essentials: Tone, pitch variation, body language, eye contact, gesture, posture, haptics, kinesics, etc. 2. Debate 3. Group discussion: roles and responsibilities of speaking in a group, team etiquette, 4. Public Speaking: case studies of successful public speakers 5. Persuasive talk: speaking to convince, agreeing to disagree, zero negations 6. Interview Techniques				
Unit No.: 5	Writing Skills	Hours 15		
1. Types of writing (formal/ informal, etc.) 1.1 Formal Letter Writing 1.2 Responding to letters 1.3 Email writing and responding to the email 1.4 Other critical formal writings (Critical Essay writing, Report Writing, book/ movie review) 1.5 Resume Writing 2. Presentation skills 3. Creative Writing: Blog writing, social profile building.				

Textbooks

1	<i>The Handbook of Communication Skills edited by Owen Hargie</i>
2	Business English by N Gupta et al 2003

Reference Books / Journal Articles / Weblink

1	Business English, Pearson, 2008.
2	<i>English and Soft Skills BY S P Dhanavel</i>
3	<i>Cambridge Soft Skills and Employability Skills by Sabina Pillai, Agna Fernandez</i>
4	Entry from Backside Only by Bino K John
5	University Grammar of English by
6	English Phonetics for Indian Students by Balsubramanian